

SIMATS ENGINEERING

Department of Computer Science & Engineering

CO5-ClassActivity

Programming in Java - File I/O Operations

Student Name	K. V. S. S. Jeswanth Babu
Register No.	192472012
Course Code	CSA0904
Course Name	Programming in Java
Assignment Topic	CO5-ClassActivity: File Creation and Writing
Question No.	Question 1

QUESTION 1: FILE CREATION AND DATA WRITING

Problem Statement:

Create a Java program to create a new file named `output.txt` and write multiple lines of structured data to it using Java File I/O (`File`, `FileWriter`, and exception handling).

Source Code (CreateAndWriteFile.java):

```
import java.io.File;
import java.io.FileWriter;
import java.io.IOException;

public class CreateAndWriteFile {
    public static void main(String[] args) {
        String fileName = "output.txt";

        try {
            // 1. Create a new file
            File file = new File(fileName);
            if (file.createNewFile()) {
                System.out.println("File created successfully: " + file.getName());
            } else {
                System.out.println("File already exists. Writing data to existing file...");
            }

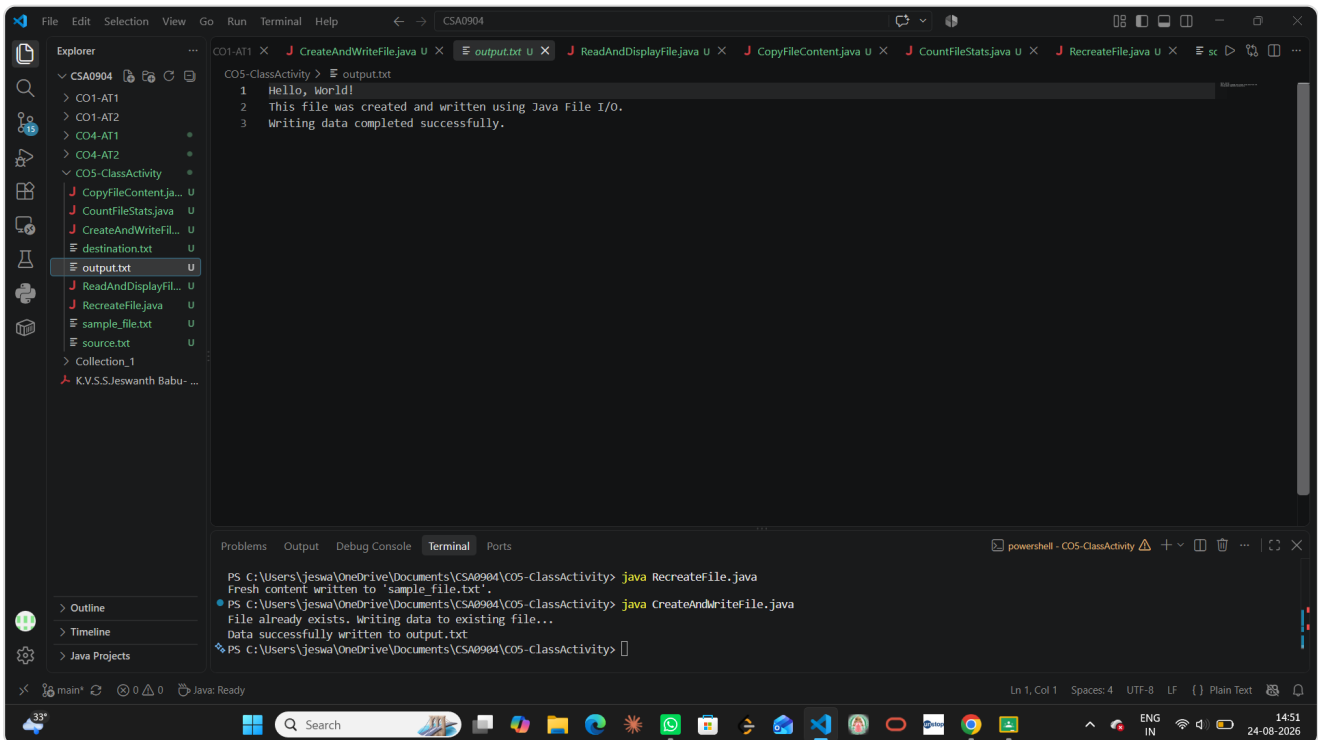
            // 2. Write data to the file using try-with-resources (auto-closes writer)
            try (FileWriter writer = new FileWriter(file)) {
                writer.write("Hello, World!\n");
                writer.write("This file was created and written using Java File I/O.\n");
                writer.write("Writing data completed successfully.");
            }

            System.out.println("Data successfully written to " + fileName);

        } catch (IOException e) {
            System.err.println("An error occurred during file operations: " + e.getMessage());
            e.printStackTrace();
        }
    }
}
```

EXECUTION & VERIFICATION OUTPUT

1. Program Execution & Terminal Output:



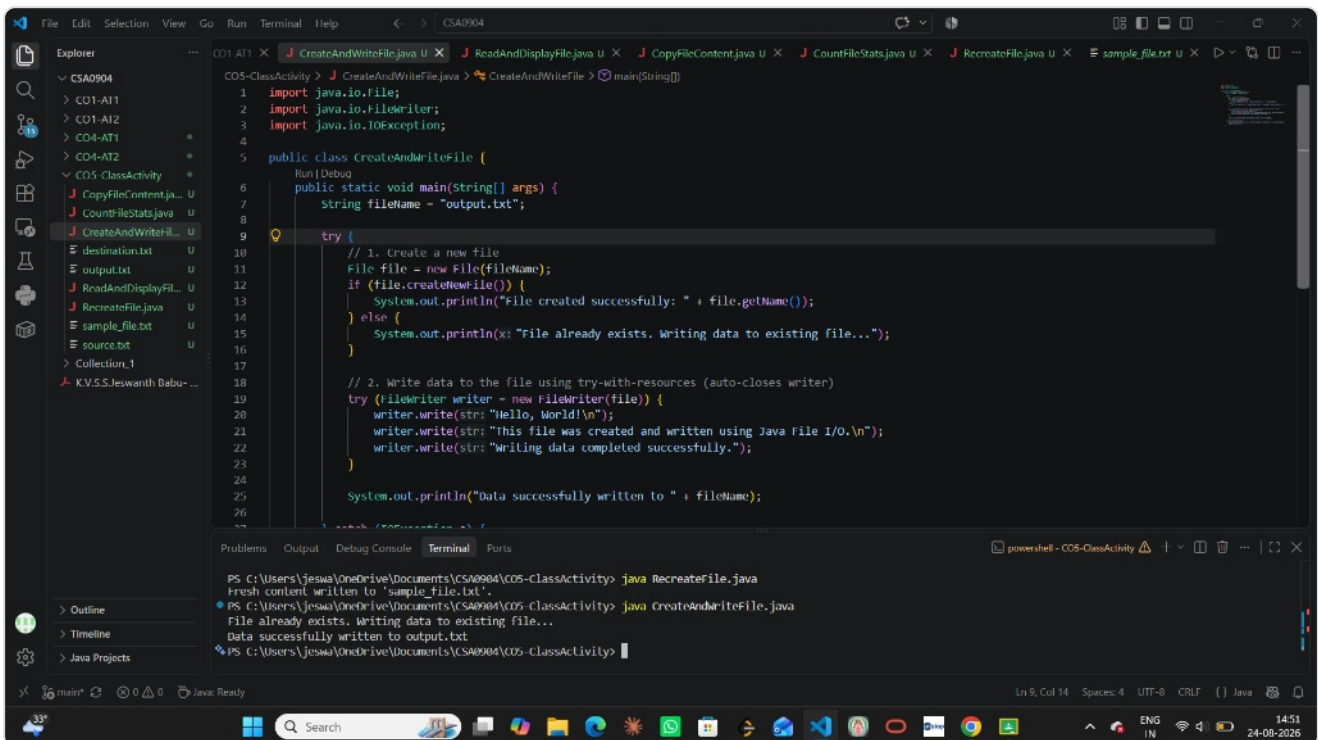
The screenshot shows the Visual Studio Code interface with the Explorer view on the left displaying a project structure. The main editor window shows the `output.txt` file with the following content:

```
1 Hello, World!
2 This file was created and written using Java File I/O.
3 Writing data completed successfully.
```

The Terminal view at the bottom shows the execution of the `RecreateFile.java` and `CreateAndWriteFile.java` programs. The output is as follows:

```
PS C:\Users\jeswa\OneDrive\Documents\CSA0904\CO5-ClassActivity> java RecreateFile.java
Fresh content written to 'sample_file.txt'.
PS C:\Users\jeswa\OneDrive\Documents\CSA0904\CO5-ClassActivity> java CreateAndWriteFile.java
File already exists. Writing data to existing file...
Data successfully written to output.txt
PS C:\Users\jeswa\OneDrive\Documents\CSA0904\CO5-ClassActivity>
```

2. Generated File Content Verification (output.txt):



The screenshot shows the Visual Studio Code interface with the Explorer view on the left displaying a project structure. The main editor window shows the `CreateAndWriteFile.java` file with the following content:

```
1 import java.io.File;
2 import java.io.FileWriter;
3 import java.io.IOException;
4
5 public class CreateAndWriteFile {
6     Run | Debug
7     public static void main(String[] args) {
8         String filename = "output.txt";
9         try {
10             // 1. Create a new file
11             File file = new File(filename);
12             if (file.createNewFile()) {
13                 System.out.println("File created successfully: " + file.getName());
14             } else {
15                 System.out.println("File already exists. Writing data to existing file...");
16             }
17
18             // 2. Write data to the file using try-with-resources (auto-closes writer)
19             try (FileWriter writer = new FileWriter(file)) {
20                 writer.write(str: "Hello, World!\n");
21                 writer.write(str: "This file was created and written using Java File I/O.\n");
22                 writer.write(str: "Writing data completed successfully.");
23             }
24
25             System.out.println("Data successfully written to " + filename);
26         } catch (IOException e) {
27             e.printStackTrace();
28         }
29     }
30 }
```

The Terminal view at the bottom shows the execution of the `RecreateFile.java` and `CreateAndWriteFile.java` programs. The output is as follows:

```
PS C:\Users\jeswa\OneDrive\Documents\CSA0904\CO5-ClassActivity> java RecreateFile.java
Fresh content written to 'sample_file.txt'.
PS C:\Users\jeswa\OneDrive\Documents\CSA0904\CO5-ClassActivity> java CreateAndWriteFile.java
File already exists. Writing data to existing file...
Data successfully written to output.txt
PS C:\Users\jeswa\OneDrive\Documents\CSA0904\CO5-ClassActivity>
```